

Coding Academy

DESIGN EMBEDDED SYSTEM HARDWARE

Term III- Exercises for the Review of concepts

Names:

=====

1. REMEMBERING LEVEL

i) Multiple choice

1. What does PCB stand for?

- A. Printed Cable Board
- B. Printed Circuit Board
- C. Power Control Board
- D. Programmed Circuit Base

2. Which PCB type has conductive layers on only one side?

- A. Multilayer PCB
- B. Flexible PCB
- C. Single-sided PCB
- D. Double-sided PCB

3. Which material is commonly used as the base substrate of PCBs?

- A. Copper
- B. Silicon
- C. FR-4
- D. Aluminum foil

4. What is the purpose of a solder mask?

- A. Increase PCB thickness
 - B. Protect copper traces
 - C. Mount components
 - D. Cool the PCB
-



5. Which technology mounts components directly on the PCB surface?

- A. THT
 - B. SMT
 - C. PTH
 - D. DIP
-

6. Which EDA tool is cloud-based according to the presentation?

- A. KiCad
 - B. Proteus
 - C. EasyEDA
 - D. OrCAD
-

7. What is the conductive path on a PCB called?

- A. Via
 - B. Pad
 - C. Track
 - D. Footprint
-

8. Which via passes through the entire PCB?

- A. Blind via
 - B. Buried via
 - C. Through-hole via
 - D. VIP via
-

9. What does EDA stand for?

- A. Electronic Design Automation
 - B. Electrical Device Assembly
 - C. Electronic Data Analysis
 - D. Electrical Design Application
-

10. Which IPC standard is related to PCB trace width?

- A. IPC-6012
- B. IPC-2221
- C. IPC-4761

D. IPC-7351

ii) Open questions

1. What does PCB stand for?

Answer:

2. What is the difference between a PCB and a PWB?

Answer:

3. Name the three PCB classifications based on the number of layers.

Answer:

4. What is the function of a via in PCB design?

Answer:

5. Define SMT.

Answer:

6. What material is commonly used as the PCB substrate?

Answer:

7. What does EDA stand for?

Answer:

8. Name two free PCB design software tools mentioned in the presentation.

Answer:

9. What is the purpose of a solder mask?

Answer:

10. What IPC standard is used for PCB trace width calculations?

Answer:



2. UNDERSTANDING LEVEL

i) Multiple choice

1. Why are multilayer PCBs suitable for high-speed applications?

- A. They are cheaper
- B. They have fewer components
- C. They provide higher signal density
- D. They use no vias

2. Why are wider traces used for power lines?

- A. To reduce board size
- B. To increase resistance
- C. To carry higher current safely
- D. To improve color appearance

3. Why should 90-degree trace bends be avoided?

- A. They increase EMI and signal reflection
- B. They reduce copper usage
- C. They simplify routing
- D. They cool the PCB

4. What is the role of vias in PCB design?

- A. Decorating the PCB
- B. Connecting different PCB layers
- C. Measuring voltage
- D. Cooling components

5. Why is FR-4 widely used in PCBs?

- A. It is transparent
- B. It is flexible only
- C. It provides strength and affordability



D. It is purely metallic

6. Why are differential pairs routed together?

- A. To reduce manufacturing cost
 - B. To maintain signal integrity
 - C. To reduce soldering
 - D. To improve silkscreen printing
-

7. What is the purpose of thermal reliefs?

- A. Increase PCB weight
 - B. Improve soldering to planes
 - C. Reduce copper thickness
 - D. Increase voltage
-

8. Why should signal traces remain over ground planes?

- A. To increase EMI
 - B. To minimize electromagnetic interference
 - C. To reduce board color
 - D. To increase routing angles
-

9. Why are SMT components preferred in compact electronics?

- A. They are larger
 - B. They require holes
 - C. They allow high component density
 - D. They are manually soldered only
-

10. Why is DRC important in PCB design?

- A. It has many mineral resources
 - B. It checks rule violations before manufacturing
 - C. It powers the PCB
 - D. It removes solder mask
-

ii) Open questions

1. What are the different materials used for the base (substrate) according to PCB use?

Answer:

2. What are the different names of vias used in multi-layer PCB and what layers do they connect respectively?

Answer:

3. Explain why multilayer PCBs are preferred in high-speed applications.

Answer:

4. Why are wider traces used for power lines in PCB routing?

Answer:

5. Explain the difference between THT and SMT technologies.

Answer:

6. Why should right-angle routing be avoided in PCB design?

Answer:

7. Explain the purpose of thermal reliefs in PCB layouts.

Answer:

8. Describe the role of silkscreen on a PCB.

Answer:

9. Why are differential pairs important in high-speed PCB design?

Answer:

10. Explain the importance of Design Rule Check (DRC).

Answer:

3. APPLYING LEVEL

i) Multiple choice



1. A PCB designer wants to reduce crosstalk between traces. Which rule should be applied?

- A. 20H Rule
 - B. 3W Rule
 - C. Thermal Rule
 - D. IPC-6012 Rule
-

2. A high-current motor line is overheating on a PCB. What should the designer do?

- A. Reduce trace width
 - B. Remove solder mask
 - C. Increase trace width
 - D. Use thinner copper
-

3. Which package is best for breadboard prototyping?

- A. BGA
 - B. QFN
 - C. DIP
 - D. SOIC
-

4. Which PCB type is most suitable for wearable electronics?

- A. Rigid PCB
 - B. Flexible PCB
 - C. Single-layer PCB
 - D. Metal-core PCB
-

5. A designer needs excellent thermal dissipation for a processor. Which package is most appropriate?

- A. DIP
 - B. BGA
 - C. SIP
 - D. Axial package
-

6. If a signal trace causes EMI problems, what should be done?

- A. Use sharp bends
- B. Keep traces short



- C. Add dangling lines
 - D. Remove grounding
-

7. Which software would be most suitable for quick online PCB prototyping?

- A. EasyEDA
 - B. MATLAB
 - C. AutoCAD
 - D. Multisim only
-

8. A designer needs a PCB for high-frequency circuits. Which material is appropriate?

- A. Paper substrate
 - B. PTFE (Teflon)
 - C. Wood
 - D. Plastic tape
-

9. Which routing practice helps reduce interference?

- A. Use 90° corners
 - B. Use self-loop routing
 - C. Separate analog and digital sections
 - D. Increase dangling traces
-

10. A PCB requires direct mounting of tiny components without drilled holes. Which technology should be selected?

- A. THT
 - B. SMT
 - C. PTH
 - D. SIP
-

11. What is a key characteristic of multi-layer PCBs compared to single-layer PCBs?

- A. Multi-layer PCBs have more layers of copper and insulation, allowing for more complex circuits.
- B. Multi-layer PCBs are only used for high-frequency applications.
- C. Multi-layer PCBs are more cost-effective than single-layer PCBs.
- D. Multi-layer PCBs have a single layer of copper with no insulation.

ii) Open questions

1. A PCB carries a high current motor supply line. Which routing rule should be applied to the trace width?

Answer:

2. Which package type would be suitable for breadboard prototyping: DIP or SOP?

Answer:

3. A designer wants better thermal dissipation for a processor. Which package among PGA, BGA, or LGA is most appropriate?

Answer:

4. Apply the 3W rule to determine spacing if the trace width is 2 mm.

Answer:

5. A circuit requires flexible movement. Which PCB material type should be selected?

Answer:

6. A PCB has strong EMI problems. Which grounding practice should be applied?

Answer:

7. Applying the IPC-2221 standard to design PCB tracks:

A PCB is designed to withstand a current of 2 A. If we want to limit temperature rise during operation at 10°C, what size of the cross-section is needed for the tracks:

(i) For Outer layer:

A. in mils²

B. in mm²

Answer:

(ii) For Inner layer:

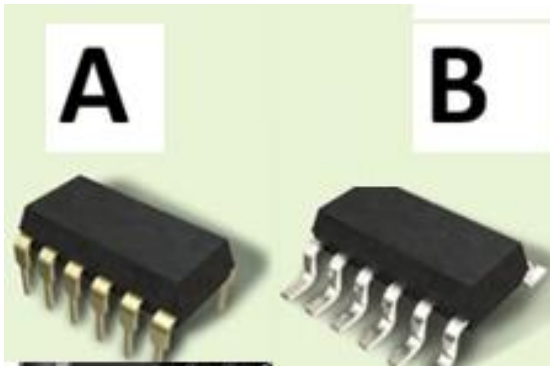
A. in mils²

B. in mm²

Answer:

(iii) What is the trace width in mils (use a standard copper for PCB) for internal and external layers.

Answer:



8.

(i) Which one is DIP and Which one is SOP?

(ii) What is DIP and SOP in full?